IDENTICAL COMPLETE AXIS SETS DO NOT IMPLY AMBIENT COMMENSURABILITY OF KLEINIAN GROUPS

ARGUS OPEN RESEARCH PROJECT

Abstract. Question 5.1 of Bestvina's problem list asks whether two finitely generated Kleinian groups in $\mathrm{PSL}_2(\mathbb{C})$ sharing the same set of loxodromic axes must be commensurable. Following Xie and Jiang (2008), we record the elementary counterexample given by two cyclic loxodromic groups with multiplicatively independent translation lengths along the same geodesic.

1. The Elementary Counterexample

Let $\ell \subset \mathbb{H}^3$ be the vertical geodesic connecting 0 and ∞ in the upper half-space model, and define

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix} \in \mathrm{PSL}_2(\mathbb{C}).$$

Proposition 1.1. *The cyclic subgroups $G_1 = \langle A \rangle$ and $G_2 = \langle B \rangle$ are finitely generated discrete subgroups of $\mathrm{PSL}_2(\mathbb{C})$ whose nonidentity elements all have the same axis $\mathrm{Ax}(G_1) = \mathrm{Ax}(G_2) = \{\ell\}$, yet $G_1 \cap G_2 = \{I\}$ because $4^m = 9^n$ in $\mathbb{Q}^\times$ forces $m = n = 0$. Hence $[G_1 : G_1 \cap G_2] = [G_2 : G_1 \cap G_2] = \infty$, and G_1, G_2 are not commensurable in $\mathrm{PSL}_2(\mathbb{C})$.*

Remark 1.2. Both G_1 and G_2 are elementary and $\langle G_1, G_2 \rangle$ is indiscrete; the non-elementary Kleinian version holds by results of Long– Reid and Xie–Jiang.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1100501 in the domain of Kleinian groups and hyperbolic geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *known counterexample reconstruction* deciding the scope: unrestricted ambient commensurability implication.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- nonelementary groups.
- jointly discrete groups.

References

[1] Primary problem list and source literature for Result 1100501 (Kleinian groups and hyperbolic geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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